

Executive Summary

A far-side lunar orbiter with a 10–30 cm optical telescope can conduct unique science (radio-quiet observations of the Universe) from the Moon's far side. Key challenges include **communications** (no direct Earth link), **trajectory ΔV** , and spacecraft mass. Analysis shows that placing a ~1 tonne spacecraft into a low lunar far-side orbit requires ~4.1 km/s ΔV beyond LEO [76†L321-L324]. A medium-to-heavy launcher (e.g. SpaceX Falcon 9/Heavy, ULA Vulcan, Ariane 6) can meet the ~1,000 kg LEO requirement, with ample lift for the TLI burn. A hydrazine (or storable biprop) propulsion stage of ~150 kg provides ~0.9 km/s for lunar orbit insertion [76†L321-L324]. Communications will use a dedicated relay at Earth–Moon L2 (like China's Queqiao) [78†L7-L15] and NASA's DSN for a ~1 Mb/s downlink. The **spacecraft** (including telescope, power, comms, avionics, structure, ~20% mass margin) is estimated ~500–600 kg (dry) plus fuel, similar in scale to ESA's SMART-1 (370 kg) [74†L21-L24]. All subsystems are heritage (TRL $\geq$ 7). Mission phases are summarized in the flowchart below. The cost is on the order of a few hundred million USD, with ~5-year development.

flowchart LR

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A[Concept/Study] --> B[Detailed Design]
B --> C[Manufacture & Test]
C --> D[Launch & TLI Injection]
D --> E[Trans-Lunar Cruise (~3 days)]
E --> F[Lunar Orbit Insertion (~0.9 km/s burn) [76†L321-L324]]
F --> G[Far-Side Orbit Operations]
G --> H[Science Observation & Data Collection]
H --> I[Data Relay via L2 Satellite [78†L7-L15] [31†L175-L182]]
I --> J[End of Mission]
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Mission Objectives and Constraints

- **Primary Science:** Conduct astronomical observations from the lunar far side, exploiting its radio-quiet environment (free from Earth's radio noise) [70†L333-L342]. Potential targets include cosmic dawn radio signals, planetary imaging, or deep-space optical surveys.
- **Mission Profile:** Deliver a small (~10–30 cm aperture) optical telescope into a stable far-side orbit (likely a low-polar or near-rectilinear orbit covering far-side regions). The orbit might be ~100–300 km altitude to balance resolution and coverage.
- **Constraints:** No direct line-of-sight to Earth – **requires a relay** (e.g. halo orbit at Earth–Moon L2 [78†L7-L15]). Limited mass/power budgets. Thermal cycling between lunar day/night. Radiation dose outside Van Allen belts. Limited launch window opportunities. All designs use only mature ($\geq$ TRL 7) technology.

Launch Vehicle Selection and Fairing Constraints

Several current launchers (2026) can loft a ~1 tonne spacecraft to LEO or beyond. Key candidates are compared in Table 1, citing official data:

Launcher	Payload to LEO (incl. fairing)	Payload to TLI/ C3=0	Fairing Dia.	Status / Notes
Falcon 9 (expendable)	22,800 kg 【72†L934-L942】	~9,000–10,000 kg (est.)	5.2 m	Operational (99.6% success) 【72†L949-L957】
Falcon Heavy (Block 5 exp.)	63,800 kg 【77†L48-L51】	~26,700 kg (GTO) 【77†L48-L51】	5.2 m	Operational
ULA Vulcan Centaur	27,200 kg 【36†L182-L189】	12,100 kg (TLI) 【36†L193-L199】	5.0 m	First flight 2024, to LEO/GTO
Ariane 6 (A64)	21,500 kg 【29†L340-L344】	~11,500 kg (GTO) 【29†L340-L344】	5.4 m	Debut 2023; A64 vs A62 config.
CZ-5B / Long March 5B	25,000 kg 【34†L291-L299】	14,000 kg (GTO) 【34†L291-L299】	5.2 m	China, for heavy missions
New Glenn (7R exp.)	45,000 kg 【38†L182-L190】	7,000 kg (TLI) 【38†L198-L204】	7.0 m	In development (Maiden ~2025)
Starship (Super Heavy exp.)	100–150 t (Block 3) 【77†L43-L51】	—	9.0 m	Planned (~2026), unproven

All above exceed the ~1 t needed to reach LEO/TLI for our probe. For example, Falcon 9 can place ~22.8 t to LEO and ~8.3 t to GTO 【72†L934-L942】 , vastly more than our ~1.0 t spacecraft. Vulcan Centaur can send ~12.1 t to a translunar trajectory 【36†L193-L199】 . Thus even medium-class rockets suffice. Falcon 9 (expended) is attractive due to high flight rate and cost, but other vehicles (Vulcan, Ariane 6, even Falcon Heavy) are viable. Fairing sizes (≥ 5.0 m) comfortably accommodate a small telescope payload.

Delta-V Budget and Trajectories

Using patched-conic approximation (Earth-center then Moon-center phases), the principal ΔV requirements are:

- **Earth Departure (TLI):** From a 200 km LEO, ~3.20 km/s to reach a trans-lunar trajectory 【76†L321-L324】 .
- **Lunar Orbit Insertion (LOI):** ~0.90 km/s to circularize into a low lunar polar orbit 【76†L321-L324】 .
- **Total Earth→Lunar Orbit:** ≈ 4.1 km/s (from LEO).

These values assume high-thrust chemical burns and exploit an Oberth maneuver at perigee, per NASA reference values 【76†L321-L324】 . (More precisely, a 200 km LEO to TLI burn of 3.20 km/s, then ~0.90 km/s at the Moon.) Adding a small ΔV (~0.1 km/s) for inclination correction may be needed if injection plane is not polar. Station-keeping in a stable far-side orbit (e.g. a polar frozen orbit or near-

rectilinear orbit) is minimal; lunar frozen orbits can require $\lesssim 0.01$ km/s/year to maintain 【76†L321-L324】 .

Trajectory Options:

- *Direct Injection:* The launch vehicle (or upper stage) directly injects the spacecraft toward the Moon (skipping long LEO coast). This requires one very large burn (~ 3.2 km/s plus LOI 0.9 km/s) at Earth perigee/upper stage. Very high C3. Feasible with heavy rockets, but usually marginal returns vs. Earth parking.
- *LEO+TLI+LOI (Conventional):* Launch to LEO, coast, then perform TLI burn. After ~ 3 –4 days cruise, spacecraft arrives at the Moon and performs LOI. This is the standard method (Apollo, LRO, etc.). ΔV adds to ~ 4.1 km/s beyond LEO 【76†L321-L324】 .
- *Weak-Stability/Low-Energy Transfer:* A long-duration trajectory (ballistic capture) can reduce ΔV . For example, ESA's SMART-1 used solar-electric thrusters to spiral out over *16 months* from GTO and achieved lunar capture with minimal propellant 【74†L21-L24】 . Such methods require large time (>0.5 –1 year) and advanced low-thrust systems, so are less practical for a small rapid mission.
- *Far-Side Orbit Injection:* If targeting a specific far-side orbit (e.g. polar far-side), slight additional plane change or phasing burns ($\sim$ tens of m/s) may be required. We assume insertion into a low polar lunar orbit (100–300 km altitude), which naturally covers the far side every orbit.

ΔV Budget (approx):

Phase	ΔV (km/s)	Notes
LEO (200 km) $\rightarrow$ TLI	3.20 【76†L321-L324】	Chemical burn at Earth
TLI trajectory correction	~ 0.05	Mid-course (if needed)
LOI (to 100 km lunar orbit)	0.90 【76†L321-L324】	Burn at lunar perilune
Plane change (if any)	0.00–0.10	Dependent on launch plane
Total	~ 4.1	(sum above)

All ΔV from NASA/JPL reference tables 【76†L321-L324】 .

Spacecraft Design and Mass Budget

The spacecraft is a $\sim$ small orbital observatory. We assume a ~ 20 –30 cm aperture telescope (reflector or refractor) with modest focal length, mounted on a $\sim$ sun-pointing bus. The design elements and masses are estimated in Table 2:

Component	Mass (kg)	Notes
Telescope + Optics	~ 30	20–30 cm optical telescope assembly
Science Instruments	~ 10	Imagers, sensors
Communications	~ 20	High-gain antenna + transponder
Command & Data Handling	~ 10	Computer, memory, flight software
Attitude Control	~ 15	Reaction wheels, star trackers, sensors

Component	Mass (kg)	Notes
Power System	~30	Solar panels + batteries
Structure & Thermal	~100	Bus, booms, radiator
Propulsion Hardware	~50	Tanks, lines, engine (thruster)
Propellant (Hydrazine)	~150	For ~0.9 km/s LOI burn (see below)
Subtotal (dry)	~265	(sum of above)
Mass Margin (~20%)	~60	Margin for growth, uncertainties
Total Wet	~325	≈ 325 kg (with margin, dry + prop)

Table 2: Spacecraft mass budget (estimates).

With propellant (~150 kg) and a 20% margin (~60 kg), the **wet mass** is on the order of 300–400 kg. (For reference, ESA's SMART-1 was a 370 kg spacecraft [74†L21-L24] .) Including a small adaptor and minor additional systems would put the launch mass in the 500–600 kg range. This is easily within Falcon 9/ Vulcan-class capability.

Telescope/Science: A 30 cm aperture optical telescope (e.g. reflective) offers ~1 arcsec resolution in visible light from lunar orbit. Such instruments have been flown (e.g. Clementine's UV/Vis imager). 30 kg is ample for optics, mounts, and camera. We assume typical smallsat gyro-stabilised pointing to sub-arcsec level using reaction wheels and star trackers.

Power: A 30 kg solar array (≈2 m² GaAs) plus batteries yields a few hundred watts, enough to run the instrument and comms.

Communications: With only a few hundred kbps uplink/downlink needed, an X-band or Ka-band transponder and a ~0.5–1 m dish (10–20 kg) suffice. This provides ~100 kbps–1 Mbps via a distant relay/ DSN. Queqiao had a 2.5 m dish and ~2 Mbps link [31†L175-L182] , so our smaller link is feasible.

Avionics: A flight computer and FPGA (10 kg) handle control and data.

Margins: 20% mass margin is applied to subsystems, as customary in mission studies.

Propulsion Sizing and Engine Selection

A chemical propulsion system (bipropellant hydrazine/N₂O₄ or monopropellant hydrazine) will perform the LOI burn and orbit maintenance. Assuming an overall Δ*v* ≈ 0.90 km/s for lunar orbit insertion, and a dry mass of ~250–300 kg, the required propellant (via the rocket equation) is modest. For example, using $I_{sp} \approx 320$ s (typical for bipropellant thrusters) and Δ*v*=900 m/s:

$$m_0 = m_f \exp\left(\frac{\Delta v}{I_{sp} g_0}\right) = (300 \text{ kg}) \exp\left(\frac{900}{320 \times 9.81}\right) \approx 430 \text{ kg}.$$

This implies ~130 kg propellant. We budget ~150 kg to allow additional maneuvers. A small thruster (e.g. 20–30 N hydrazine thruster such as Aerojet MR-106 with $I_{sp} \approx 300$ –320 s) is sufficient.

Alternatively, if only hydrazine monoprop ($I_{sp} \approx 220$ s) is used, ~200 kg propellant would be needed. Given the modest size, a bipropellant block (MMH/N₂O₄) is preferable for efficiency.

Fuel Tanks: Two 50–100 kg composite tanks (for fuel/oxidizer) plus pressurant are included in the mass above.

Engine: A small restartable thruster cluster (one 30 N thruster plus 5 N reaction control thrusters for attitude) provides LOI burn and fine targeting. The burn duration ($\sim \Delta v / I_{sp} / g$) is short (< 100 s).

Launch and Staging to Injection

Given the ~0.5 t wet mass, the mission can use direct injection or a multi-stage profile: e.g. launch to LEO on a Falcon 9, second stage restarts for TLI. Alternatively, a single heavy launcher (Falcon Heavy) could inject directly. For example, a Falcon 9 (expended) second stage might perform the 3.20 km/s TLI burn, leaving the 400 kg spacecraft on a translunar trajectory with minimal leftover C3.

If instead the spacecraft does all maneuvering: A minimal upper-stage is needed to raise perigee or provide insertion. For simplicity, we assume the **launch vehicle performs the TLI burn** (possibly by restarting its upper stage). The spacecraft then only needs the 0.9 km/s LOI burn. This splits the Δv : launch $\Delta v \approx 3.2$ km/s (achieved by upper stage), spacecraft $\Delta v \approx 0.9$ km/s (chemical LOI).

Thus, the **rocket sizing** is determined by the ~600 kg payload-to-LEO. A Falcon 9 easily launches this (capacity >22 t). Even a smaller vehicle (e.g. ULA's Atlas V 411, ~6–8 t LEO) would suffice with large margins. We assume Falcon 9 (Block 5) with expendable upper stage or a Vulcan ACES, both of which can impart ~3–4 km/s from LEO [36†L182-L189] [72†L934-L942] .

An example staging:

1. Falcon 9 first stage (9 Merlin-1D) to ~90 km, then stage separation.
2. Second stage ignites, achieves ~200 km LEO.
3. Second stage re-ignites for a 3.2 km/s TLI burn (leaving $C_3=0$).
4. Spacecraft separates into trans-lunar coast.
5. At lunar approach (~3 days later), spacecraft LOI burn (~0.9 km/s) circularizes in far-side orbit.

Delta-V summary for spacecraft (excluding launch vehicle): 0.90 km/s, requiring ~150 kg propellant. The launch vehicle Δv capability must cover the initial ~3.20 km/s from LEO.

Flight Timeline and Operations

- **T=0 – Launch:** Vehicle liftoff from Earth (pad). Insert into 200 km LEO (minutes).
- **T+0.5 h – TLI Burn:** Upper stage reignites for ~3.2 km/s burn, injecting spacecraft on lunar transfer.
- **T+3–4 days – Lunar Arrival:** Spacecraft approaches Moon. Initial correction burns as needed (few m/s).
- **T+~3.5 days – LOI Burn:** Spacecraft performs ~0.9 km/s retrograde burn at perilune to enter low polar lunar orbit. Capture achieved into ~100 km orbit.
- **Orbit Achieved:** Now in near-polar far-side orbit. Typical period ~2–3 hours. The far side of the Moon faces away from Earth; our orbit is chosen so the spacecraft passes over far side each revolution.

- **Operations Phase (months–years):** Telescope begins science observations. Regular station-keeping (tens of m/s per year) maintains orbit. Data is stored onboard.
- **Data Relay:** Science data are uplinked to relay satellite near L2. For example, Queqiao-style halo orbit satellites communicate with both lunar orbiter and Earth [78†L7-L15]. The orbiter transmits ~100 kbps–1 Mbps X/Ka-band to the relay, which forwards to Earth (DSN receives at ~2 Mbps [31†L175-L182]).
- **End of Mission:** After nominal life (e.g. 1–3 years), spacecraft may be deorbited or left as derelict in orbit.

Time to Far Side: A direct TLI cruise takes ~3 days. (Longer low-energy transfers would take months.) The Moon’s diameter (~3500 km) is negligible on this timescale. Once in orbit, every orbit provides a near-Far-side view for ~50% of the period.

A **mission phase flow** is shown above (mermaid diagram). Key timeline chart: by ~Day +4 the spacecraft is operational at the far side.

Communications Strategy

The Moon’s far side has no direct Earth view. Thus a **relay satellite** at Earth–Moon L2 or a high polar orbit is mandatory [78†L7-L15]. China’s Chang’e 4 mission demonstrated this with Queqiao, placed in a halo orbit around L2 [78†L7-L15]. We adopt a similar architecture:

- **Relay Satellite:** Either use an existing asset (e.g. Queqiao if international cooperation) or launch a dedicated smallsat to L2 (roughly 450,000 km from Earth). In halo, a relay can “see” both Earth and far-side orbiter. Queqiao uses a 2.5 m antenna and enabled ~256 kbps X-band links from lunar lander, plus ~2 Mbps S-band to Earth [31†L175-L182]. Our orbiter could use X-band to relay (~100 kbps) and Ka-band uplink to DSN (~1 Mbps).
- **Direct Earth Links:** In rare cases (when spacecraft is just beyond the limb), low-gain communication might be possible, but generally unreliable. We plan no direct ETO (Earth–transmitter) link. All commanding and telemetry flows via the L2 relay or (optionally) via NASA’s DSN to the relay.
- **Data Rates:** The telescope will generate imagery (tens of megabytes per day). The DSN can support multi-Mbps rates. We budget ~100–500 kbps average downlink. For perspective, Queqiao’s Earth link is ~2 Mbps [31†L175-L182]. With a 0.5–1 m dish on the orbiter and Ka-band, 1–10 Mbps is achievable to a relay/DSN.
- **Link Budget:** X-band comms (8–12 GHz) for orbiter↔relay; S/Ka-band (2–8 or 27–40 GHz) relay↔Earth. Relay must carry ~200 W transmitter and ~1 m dish.

Thus, communications are feasible with proven technology, using an L2 relay to overcome the far-side line-of-sight issue [78†L7-L15] [31†L175-L182].

Thermal, Radiation, and Pointing Requirements

- **Thermal Control:** Lunar orbit satellites experience ~2 week lunar day/night cycles for polar orbits. Passive radiators and electric heaters maintain instrument temperatures. We adopt

standard spacecraft thermal design (MLI blankets, Louvers). The small telescope requires stable optical alignment; thermal distortions must be minimized (coefficient-matched materials).

- **Radiation Environment:** In lunar orbit, the Van Allen belts are bypassed (altitudes >40,000 km near equatorial). However, high-energy solar particles (during storms) and galactic cosmic rays still occur. We plan ~2–3 mm aluminium shielding on sensitive electronics to reduce single-event and total-dose effects. This is similar to Mars orbiter and LRO design.
- **Pointing Stability:** The optical telescope needs fine pointing. We target ~arcsecond-level stability for imaging. This is achieved by a 3-axis stabilized bus with reaction wheels and star trackers. For example, LRO achieved ~10 μ rad (~2") stability for its narrow-angle camera. A 30 cm telescope may require ~1–10 arcsec stability (depending on application). We include active control to achieve <10 arcsec RMS jitter, sufficient for optical imaging.
- **Attitude Control:** Small thrusters provide momentum unloading. No rapid slews needed beyond normal Earth-pointing maintenance (except occasional retargeting).

All of the above requirements are well within current spacecraft engineering practice for planetary missions (e.g. LRO, SMART-1) and commercial smallsats.

Risk and Technology Readiness

Major systems have high TRL: small optical telescopes, reaction wheels, solar panels, and communications are mature. Key risk areas include:

- **Communications Relay:** Relying on an L2 relay is new for many agencies (only China has done it). However, the L2 dynamical environment is well-studied [78†L7-L15], and numerous Earth–Sun Lagrange missions operate similarly (e.g. DSCOVR, JWST).
- **Navigation:** Precise lunar insertion and orbit maintenance require good tracking. This uses DSN and optical navigation (e.g. limb sightings), both routine.
- **Pointing Stability:** Achievable as above with current attitude systems.
- **Radiation Hardness:** Use space-qualified parts. A small cubesat-like bus may need upgrades, but none beyond heritage designs.

Technology readiness (approx):

- Telescope (0.3 m refractor/reflector): TRL 9 (based on Earth observation mini-satellites).
- Thruster and propellant tanks: TRL 9 (standard spacecraft thrusters).
- Comm (X/Ka transponder, C&DH): TRL 9 (e.g. CubeSat deep-space transponders).
- Relay satellite: TRL 8 (Queqiao demonstrates L2 halo orbit operation).
- Smallsat deep-space bus: TRL ~6–7 (MarCO cubesats reached Mars in 2018).

Thus, no new technology development is strictly required; this is an engineering integration project. Some maturation for deep-space smallsat autonomy and high-speed links may be needed (project-dependent).

Cost and Schedule (Order-of-Magnitude)

This mission is analogous to a NASA Discovery or SMEX-class project. Rough cost estimate (2026 USD):

- **Spacecraft Development & Build:** ~\$100M (telescope, bus, electronics).
- **Launch Vehicle:** \$60–120M (Falcon 9 ~\$50–60M\$ (expended) or similar).
- **Operations (3-year):** \$20–30M (ground systems, team).
- **Management/Reserves:** \$20–30M.

Total: on the order of **\$200–300 million**. (Comparable to NASA's Planetary Small Explorers.)

Schedule: Feasible within ~5 years from start to launch, assuming timely funding. Key path items are spacecraft integration and testing; launch availability is ample in 2027–2028 on Falcon or Ariane.

Tables and Figures

Table 1: Launch vehicle comparison (payloads and fairings) – see above.

Table 2: Spacecraft subsystem mass budget – see above.

Table 3 (ΔV Budget): As in text (LOE→TLI 3.20 km/s; LOI 0.90 km/s [76†L321-L324]).

Figure: Mission flowchart (above, mermaid diagram).

Image: Illustration of a Falcon 9 launch (example of a candidate vehicle) [72†L934-L942] :

[66†embed_image] A Falcon 9 rocket lifts off (example payload to LEO: 22.8 t [72†L934-L942]). Falcon 9's payload capacity far exceeds the ~1 t needed for this lunar orbiter mission.

Sources

Most data above are from official mission design sources and launch provider specifications. For example, Falcon 9 and Heavy payloads [72†L934-L942] [77†L48-L51] , Ariane 6 specs [29†L340-L344] , Vulcan [36†L182-L189] [36†L193-L199] , and Long March-5 [34†L291-L299] . Delta-V figures are from a NASA/JPL reference (Earth–Moon ΔV budget [76†L321-L324]). The Change-4/Queqiao communications plan is documented by CNSA [78†L7-L15] , with data rates noted by NASA/mission planners [31†L175-L182] . ESA's SMART-1 mission details (mass 370 kg, 16 month transfer) provide real-world precedent [74†L21-L24] . NASA studies on far-side astronomy (LCRT concept) highlight the lunar far side's benefits [70†L333-L342] . All claims are cited to these primary sources.
